



Math 111: Linear Algebra

First Exam, 4 March 2019

Duration: 90 minutes

- In questions 1-7, show all your work.
- In question 8, select the correct answer (no partial credit is given).
- Calculators and communication devices are not allowed in the examination room.

Name: _____

Student ID: _____

Section Number: _____ Serial Number: _____

Q1 (10 pts)	
Q2 (10 pts)	
Q3 (10 pts)	
Q4 (10 pts)	
Q5 (10 pts)	
Q6 (15 pts)	
Q7 (15 pts)	
Q8 (20 pts)	
Total	

1. [10 pts.] Show that if $p(x) = x^2 - (a + d)x + (ad - bc)$ and $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, then $p(A) = O$.
2. [10 pts.] Let $(A^T)^{-1} = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ -1 & 0 & 2 \end{bmatrix}$. Solve the linear system $AX = \begin{bmatrix} 4 \\ 4 \\ 4 \end{bmatrix}$.
3. [10 pts.] Let A be an $n \times n$ invertible matrix and k be a nonzero scalar. Show that kA is invertible and $(kA)^{-1} = k^{-1}A^{-1}$.
4. [10 pts.] Let $\left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & a+2 & a^2+4a+3 \end{array} \right]$ be the augmented matrix of a linear system.

Find all values of a , if any, for which the system has at least one solution.

5. [10 pts.] Let $A = \begin{bmatrix} 1 & 0 & k \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 & 2 \\ -1 & 2 & -1 \\ 0 & 1 & 1 \end{bmatrix}$. Find all values of k , if any, such that $ABA^T = O$.

6. [15 pts.] Solve the linear system

$$\begin{array}{rrcr} x_1 & & & +4x_4 & = & 4 \\ -2x_1 & +x_2 & -x_3 & -3x_4 & = & -9 \\ -2x_1 & +x_2 & & -7x_4 & = & -8. \end{array}$$

7. [10 + 5 = 15 pts.] Let $A = \begin{bmatrix} 1 & 4 & a \\ 0 & 2 & 1 \\ 2 & 8 & a^2 \end{bmatrix}$.

- (a) Find all values of a , if any, for which A^{-1} exists.
- (b) For $a = 1$, solve the system $AX = O$.

8. [$5 \times 4 = 20$ pts.] In each of the following questions, select the correct answer.

- I. Let $A = \begin{bmatrix} 2 & 3 \\ -1 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 9 \\ -3 & k \end{bmatrix}$. The value of k for which $AB = BA$ is equal:
- (A) $k = 3$
 - (B) $k = -3$
 - (C) $k = 8$
 - (D) $k = -8$
 - (E) None of the above.
- II. If A is an $n \times n$ matrix satisfying the equation $A^2 + 5A - 2I_n = O$, then A^{-1} is equal to:
- (A) $(A + 5I_n)$
 - (B) $(A - 5I_n)$
 - (C) $\frac{1}{2}(A + 5I_n)$
 - (D) $\frac{1}{2}(A - 5I_n)$
 - (E) None of the above.
- III. Let $A = \begin{bmatrix} a & 2 & d \\ e & b & -1 \\ 0 & f & c \end{bmatrix}$. The values of a, b, c, d, e and f for which A is skew-symmetric are:
- (A) $a = b = c = d = 0, e = -2, f = 1$
 - (B) $a = b = c = d = 0, e = 2, f = -1$
 - (C) $a, b, c \in \mathbb{R}, d = 0, e = 2, f = -1$
 - (D) $a, b, c \in \mathbb{R}, d = 0, e = -2, f = 1$
 - (E) None of the above.
- IV. If $A = \begin{bmatrix} 2 & 5 \\ 6 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 7 \\ 8 & 3 \end{bmatrix}$, then $\text{tr}(2A + 3B)$ is equal to:
- (A) 18
 - (B) 13
 - (C) 5
 - (D) 6
 - (E) None of the above.

1. [10 pts.] Show that if $p(x) = x^2 - (a+d)x + (ad-bc)$ and $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, then $p(A) = O$.

$$p(A) = A^2 - (a+d)A + (ad-bc)I_2 = \begin{bmatrix} a^2+bc & ab+bd \\ ac+cd & cb+d^2 \end{bmatrix} - \begin{bmatrix} a^2+ad & ab+bd \\ ac+cd & ad+d^2 \end{bmatrix} + \begin{bmatrix} ad-bc & 0 \\ 0 & ad-bc \end{bmatrix} = O.$$

2. [10 pts.] Let $(A^T)^{-1} = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ -1 & 0 & 2 \end{bmatrix}$. Solve the linear system $AX = \begin{bmatrix} 4 \\ 4 \\ 4 \end{bmatrix}$.

Since $(A^T)^{-1} = (A^{-1})^T$, $\therefore A^{-1} = \begin{bmatrix} 1 & 0 & -1 \\ 2 & 1 & 0 \\ 3 & 2 & 2 \end{bmatrix}$. The system $AX = \begin{bmatrix} 4 \\ 4 \\ 4 \end{bmatrix}$ has a unique solution,

$$X = \begin{bmatrix} 1 & 0 & -1 \\ 2 & 1 & 0 \\ 3 & 2 & 2 \end{bmatrix} \begin{bmatrix} 4 \\ 4 \\ 4 \end{bmatrix} = \begin{bmatrix} 0 \\ 12 \\ 28 \end{bmatrix}.$$

3. [10 pts.] Let A be an $n \times n$ invertible matrix and k be a nonzero scalar. Show that kA is invertible and $(kA)^{-1} = k^{-1}A^{-1}$.

We have $(kA)(k^{-1}A^{-1}) = k^{-1}(kA)A^{-1} = (k^{-1}k)AA^{-1} = (1)I = I$ and similarly $(k^{-1}A^{-1})(kA) = I$. Thus, kA is invertible and $(kA)^{-1} = k^{-1}A^{-1}$.

4. [10 pts.] Let $\left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & a+2 & a^2+4a+3 \end{array} \right]$ be the augmented matrix of a linear system.

Find all values of a , if any, for which the system has at least one solution.

From the augmented matrix $\left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & a+2 & (a+3)(a+1) \end{array} \right]$, it is clear that if $a = -2$ the system, has no solution, otherwise it has at least one solution. Thus $a \in \mathbb{R} - \{-2\}$.

5. [10 pts.] Let $A = \begin{bmatrix} 1 & 0 & k \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 & 2 \\ -1 & 2 & -1 \\ 0 & 1 & 1 \end{bmatrix}$. Find all values of k , if any, such that $ABA^T = O$.

$$ABA^T = \begin{bmatrix} 1 & 0 & k \end{bmatrix} \begin{bmatrix} 1 & 0 & 2 \\ -1 & 2 & -1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ k \end{bmatrix} = \begin{bmatrix} 1 & k & 2+k \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ k \end{bmatrix} = 1 + 2k + k^2 = (k+1)^2 = 0 \iff k = -1.$$

6. [15 pts.] Solve the linear system

$$\begin{array}{rrcr} x_1 & & & +4x_4 & = & 4 \\ -2x_1 & +x_2 & -x_3 & -3x_4 & = & -9 \\ -2x_1 & +x_2 & & -7x_4 & = & -8. \end{array}$$

Since $\left[\begin{array}{cccc|c} 1 & 0 & 0 & 4 & 4 \\ -2 & 1 & -1 & -3 & -9 \\ -2 & 1 & 0 & -7 & -8 \end{array} \right]$ is row equivalent to $\left[\begin{array}{cccc|c} 1 & 0 & 0 & 4 & 4 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & -4 & 1 \end{array} \right]$, the linear system has infinitely many solutions (x_4 is a free variable). If we let $x_4 = t$, where $t \in \mathbb{R}$, then the general solution is $x_1 = 4 - 4t$, $x_2 = -t$, $x_3 = 1 + 4t$, $x_4 = t$, $t \in \mathbb{R}$.

7. [10 + 5 = 15 pts.] Let $A = \begin{bmatrix} 1 & 4 & a \\ 0 & 2 & 1 \\ 2 & 8 & a^2 \end{bmatrix}$.

(a) Find all values of a , if any, for which A^{-1} exists.

Since $\begin{bmatrix} 1 & 4 & a \\ 0 & 2 & 1 \\ 2 & 8 & a^2 \end{bmatrix}$ is row equivalent to $\begin{bmatrix} 1 & 4 & a \\ 0 & 2 & 1 \\ 0 & 0 & a^2 - 2a \end{bmatrix}$, therefore A is nonsingular iff $a(a-2) \neq 0$.

Thus, A^{-1} exists for $a \in \mathbb{R} - \{0, 2\}$.

(b) For $a = 1$, solve the system $AX = O$.

Since for $a = 1$ the matrix A is nonsingular, therefore, $AX = O$ has only the trivial solution.

8. [$5 \times 4 = 20$ pts.] In each of the following questions, select the correct answer.

I. Let $A = \begin{bmatrix} 2 & 3 \\ -1 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 9 \\ -3 & k \end{bmatrix}$. The value of k for which $AB = BA$ is equal:

(A) $k = 3$

(B) $k = -3$

(C) $k = 8$

(D) $k = -8$

(E) None of the above.

II. If A is an $n \times n$ matrix satisfying the equation $A^2 + 5A - 2I_n = O$, then A^{-1} is equal to:

(A) $(A + 5I_n)$

(B) $(A - 5I_n)$

(C) $\frac{1}{2}(A + 5I_n)$

(D) $\frac{1}{2}(A - 5I_n)$

(E) None of the above.

III. Let $A = \begin{bmatrix} a & 2 & d \\ e & b & -1 \\ 0 & f & c \end{bmatrix}$. The values of a, b, c, d, e and f for which A is skew-symmetric are:

(A) $a = b = c = d = 0, e = -2, f = 1$

(B) $a = b = c = d = 0, e = 2, f = -1$

(C) $a, b, c \in \mathbb{R}, d = 0, e = 2, f = -1$

(D) $a, b, c \in \mathbb{R}, d = 0, e = -2, f = 1$

(E) None of the above.

IV. If $A = \begin{bmatrix} 2 & 5 \\ 6 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 7 \\ 8 & 3 \end{bmatrix}$, then $\text{tr}(2A + 3B)$ is equal to:

(A) 18

(B) 13

(C) 5

(D) 6

(E) None of the above.